

Elliptic Curves

Written by HADIOUCHE Azouaou.

Disclaimer

This course is presented by Pr. STEVENHAGEN Peter.

To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

Contents

CHAPTER 1: INTRODUCTION	1
1.1. Complex Lattices	1

Chapter 1

Introduction

The fundamental theorem of calculus $\int_a^b f(t) dt = F(b) - F(a)$ with $F' = f$ gave a really useful tool to compute integrations with anti-derivatives. In the 19th century, the computation of the arc-length of function graphs became a necessity in multiple fields, and it was being computed as $\int_a^b \sqrt{1 + f'(t)^2} dt$, which has no clear way to compute an anti-derivative. For example, computing the constant π , you have that π is computed as the arc-length of $t \mapsto \sqrt{1 - t^2} dt$, and we get

$$\pi = \int_{-1}^1 \frac{dt}{\sqrt{1 - t^2}}$$

using complex analysis, you can try to compute the integral as $z \in \mathbb{C} \mapsto \int_0^z \frac{dt}{1 - t^2}$, the integral would be path dependent by the existence of singularities at ± 1 . Another way to look at the problem, we define the Riemann surface defined as $C : X^2 + Y^2 = 1$, taking $P = (0, 1)$, a lifting of $0 \in \mathbb{C}$ to C , we have $F(z) = \int_P^z \frac{dX}{Y}$ hence we obtain a function

$$\begin{aligned} C &\rightarrow \mathbb{C}/2\pi\mathbb{Z}, \\ z &\mapsto F(z) \\ (\sin z, \cos z) &\leftrightarrow z \end{aligned}$$

which is an isomorphism, $\mathbb{C}/2\pi\mathbb{Z}$ is a group, thus we get a group structure on the Riemann surface C , which is defined as

$$\begin{aligned} (\sin x, \cos x) \oplus (\sin y, \cos y) &= (\sin(x + y), \cos(x + y)) \\ &= (\sin x \cos y + \sin y \cos x, \cos x \cos y - \sin x \sin y) \end{aligned}$$

by replacing the points $(\sin x, \cos x) = (a, b)$ and $(\sin y, \cos y) = (c, d)$ then we have the operation is defined as

$$(a, b) \oplus (c, d) = (ad + bc, bd - ac)$$

we have that $\frac{dX}{Y} = \frac{d \sin z}{\cos z} = dz$. The same process can be done on the function

$$F(z) = \int_0^z \frac{dt}{\sqrt{1 - t^4}}$$

which is called the lemniscate integral. The lemniscate is defined as given two points, Q_1, Q_2 , the set of points for which the product of distances with Q_1 and Q_2 is 1, $\{P \mid |PQ_1| \cdot |PQ_2| < 1\}$. Similar to earlier, we can take the algebraic curve, $C : X^4 + Y^2 = 1$. Representing this set of points, you will get a shape that is topologically equivalent to a torus. Again, by the same process, we have that $F(z) = \int_P^z \frac{dX}{Y}$ hence we try to compute $\frac{dX}{Y}$ to obtain an algebraic representation of the Riemann surface C , we obtain that it is isomorphic to

$$\begin{aligned} C &\rightarrow \mathbb{C}/(\omega_1\mathbb{Z} + \omega_2\mathbb{Z}) \\ z &\mapsto \int_P^z \frac{dX}{Y} \\ (P(z), P'(z)) &\leftrightarrow z \end{aligned}$$

we have that $P^4 + (P')^2 = 1$, which are called elliptic functions.

1.1. Complex Lattices

Let $\omega_1, \omega_2 \in \mathbb{C}$ linearly independent, Define a lattice $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$. Let F be the fundamental domain, $F = \{\lambda_1\omega_1 + \lambda_2\omega_2 \mid \lambda_1, \lambda_2 \in [0, 1]\}$. A complex meromorphic function f cannot be analytic since it would become constant on the complex plane hence a constant function, if f has a single singularity, then $\int_{\partial F} f(z) dz = 0$, hence you would have at least 2 singularities.

We construct the function as follows

$$\wp_\Lambda(z) = \frac{1}{z^2} + \sum_{\lambda \in \Lambda \setminus \{0\}} \left[\frac{1}{(z - \lambda)^2} - \frac{1}{\lambda^2} \right]$$

which has a double pole at every lattice point, and we obtain the definition of the Weiestrass $\wp$ function, which converges uniformly. By computing the derivative, we obtain

$$\wp'_\Lambda(z) = -2 \sum_{\lambda \in \Lambda} \frac{1}{(z - \lambda)^3}$$

which is periodic. Notice also that $\wp_\Lambda$ is an even function, and hence we can use the periodicity of the derivative to prove that $\wp_\Lambda$ is periodic. We obtain the following theorem, that $m_\Lambda = \mathbb{C}(\wp_\Lambda, \wp'_\Lambda)$, meaning, any meromorphic function on Λ can be obtained algebraically with a polynomial expression of $\wp_\Lambda$ and $\wp'_\Lambda$. To prove that, we take

$$\begin{aligned} m_\Lambda^* \rightarrow \text{Div}(\Lambda) &= \bigoplus_{t \in T} \mathbb{Z} \cdot [t] \\ f \mapsto \text{div}(f) &= \sum_{t \in T} \text{ord}_t(f) \cdot [t] \end{aligned}$$

where $T = \mathbb{C}/\Lambda$. For example we have

$$\begin{aligned} \text{div}(\wp_\Lambda) &= (a) + (-a) - 2(0) \\ \text{div}(\wp'_\Lambda) &= \left(\frac{\omega_1}{2}\right) + \left(\frac{\omega_2}{2}\right) + \left(\frac{\omega_1 + \omega_2}{2}\right) - 3(0) \end{aligned}$$

We get as a theorem that the following exact sequence

$$1 \rightarrow \mathbb{C}^* \rightarrow m(T)^* \xrightarrow{\text{div}} \text{Div}^0(T) \xrightarrow{!} T \rightarrow 0$$

where div is the divisor function, and $!$ turns the formal sum into an actual sum $\sum_{t \in T} n_t \cdot [t] \mapsto \sum_{t \in T} n_t t \in T$. We have also a theorem stating that $(\wp'_\Lambda)^2 \in \mathbb{C}(\wp_\Lambda)$ and after a computation, we have

$$(\wp'_\Lambda)^2 = 4\wp_\Lambda^3 - g_2(\Lambda)\wp_\Lambda - g_3(\Lambda)$$

Similar to the previous examples, we obtain a function

$$\begin{aligned} \mathbb{C}/\Lambda &\xrightarrow{\sim} Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3 \\ z &\mapsto [\wp_\Lambda(z) : \wp'_\Lambda(z) : 1] \\ 0 &\mapsto [0 : 1 : 0] \end{aligned}$$

where we use the projective plane to define the values of the map.